

CITED REFERENCES AND FURTHER READING:

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- Lakshmikantham, V., and Trigiante, D. 1988, *Theory of Difference Equations: Numerical Methods and Applications* (San Diego: Academic Press).[3]
- Acton, F.S. 1970, *Numerical Methods That Work*, 1990, corrected edition (Washington, DC: Mathematical Association of America), pp. 20ff.[4]
- Clenshaw, C.W. 1962, *Mathematical Tables*, vol. 5, National Physical Laboratory (London: H.M. Stationery Office).[5]
- Dahlquist, G., and Björck, A. 1974, *Numerical Methods* (Englewood Cliffs, NJ: Prentice-Hall); reprinted 2003 (New York: Dover), §4.4.3, p. 111.
- Goodwin, E.T. (ed.) 1961, *Modern Computing Methods*, 2nd ed. (New York: Philosophical Library), p. 76.
- Hockney, R.W., and Jesshope, C.R. 1988, *Parallel Computers 2: Architecture, Programming, and Algorithms* (Bristol and Philadelphia: Adam Hilger), §5.2.4 and §5.4.2.[6]

5.5 Complex Arithmetic

Since C++ has a built-in class `complex`, you can generally let the compiler and the class library take care of complex arithmetic for you. Generally, but not always. For a program with only a small number of complex operations, you may want to code these yourself, in-line. Or, you may find that your compiler is not up to snuff: It is disconcertingly common to encounter complex operations that produce overflows or underflows when both the complex operands and the complex result are perfectly representable. This occurs, we think, because software companies mistake the implementation of complex arithmetic for a completely trivial task, not requiring any particular finesse.

Actually, complex arithmetic is not *quite* trivial. Addition and subtraction are done in the obvious way, performing the operation separately on the real and imaginary parts of the operands. Multiplication can also be done in the obvious way, with four multiplications, one addition, and one subtraction:

$$(a + ib)(c + id) = (ac - bd) + i(bc + ad) \quad (5.5.1)$$

(the addition sign before the i doesn't count; it just separates the real and imaginary parts notationally). But it is sometimes faster to multiply via

$$(a + ib)(c + id) = (ac - bd) + i[(a + b)(c + d) - ac - bd] \quad (5.5.2)$$

which has only three multiplications (ac , bd , $(a + b)(c + d)$), plus two additions and three subtractions. The total operations count is higher by two, but multiplication is a slow operation on some machines.

While it is true that intermediate results in equations (5.5.1) and (5.5.2) can overflow even when the final result is representable, this happens only when the final

answer is on the edge of representability. Not so for the complex modulus, if you or your compiler is misguided enough to compute it as

$$|a + ib| = \sqrt{a^2 + b^2} \quad (\text{bad!}) \quad (5.5.3)$$

whose intermediate result will overflow if either a or b is as large as the square root of the largest representable number (e.g., 10^{19} as compared to 10^{38}). The right way to do the calculation is

$$|a + ib| = \begin{cases} |a|\sqrt{1 + (b/a)^2} & |a| \geq |b| \\ |b|\sqrt{1 + (a/b)^2} & |a| < |b| \end{cases} \quad (5.5.4)$$

Complex division should use a similar trick to prevent avoidable overflow, underflow, or loss of precision:

$$\frac{a + ib}{c + id} = \begin{cases} \frac{[a + b(d/c)] + i[b - a(d/c)]}{c + d(d/c)} & |c| \geq |d| \\ \frac{[a(c/d) + b] + i[b(c/d) - a]}{c(c/d) + d} & |c| < |d| \end{cases} \quad (5.5.5)$$

Of course you should calculate repeated subexpressions, like c/d or d/c , only once.

Complex square root is even more complicated, since we must both guard intermediate results and also enforce a chosen branch cut (here taken to be the negative real axis). To take the square root of $c + id$, first compute

$$w \equiv \begin{cases} 0 & c = d = 0 \\ \sqrt{|c|} \sqrt{\frac{1 + \sqrt{1 + (d/c)^2}}{2}} & |c| \geq |d| \\ \sqrt{|d|} \sqrt{\frac{|c/d| + \sqrt{1 + (c/d)^2}}{2}} & |c| < |d| \end{cases} \quad (5.5.6)$$

Then the answer is

$$\sqrt{c + id} = \begin{cases} 0 & w = 0 \\ w + i \left(\frac{d}{2w} \right) & w \neq 0, c \geq 0 \\ \frac{|d|}{2w} + iw & w \neq 0, c < 0, d \geq 0 \\ \frac{|d|}{2w} - iw & w \neq 0, c < 0, d < 0 \end{cases} \quad (5.5.7)$$

CITED REFERENCES AND FURTHER READING:

- Midy, P., and Yakovlev, Y. 1991, "Computing Some Elementary Functions of a Complex Variable," *Mathematics and Computers in Simulation*, vol. 33, pp. 33–49.
- Knuth, D.E. 1997, *Seminumerical Algorithms*, 3rd ed., vol. 2 of *The Art of Computer Programming* (Reading, MA: Addison-Wesley) [see solutions to exercises 4.2.1.16 and 4.6.4.41].